



Höhere Technische Bundeslehranstalt Wels

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ELECTROTECHNICAL LABORATORY

Title: **14. Rectifier**

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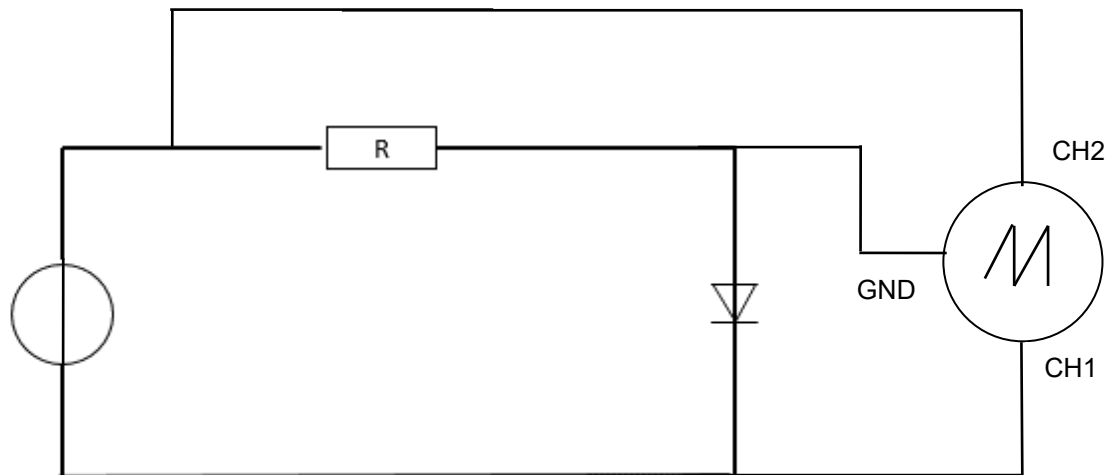
Used instruments:

Gwinstek GDS-1022 824-30-229; W1.17-225 - oscilloscope
Fluke 179 True RMS W1.17-53
Fluke 177 True RMS W1.17-51
1000 Ω plug-in resistor
100 Ω plug-in resistor
capacitors with 20nF, 47 μ F, 470 μ F
UF5408 – diode
W1.17 -705 – isolation amplifier
W1.17 -703 – isolation amplifier

Remarks:

14.1 Silicon Diode:

Preparation:



oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17-225

diode: UF5408

R_{Shunt} : 12.5 Ω resistor; $P_{\text{max}} = 4\text{W}$

$$U_{\text{Shunt_max}} = \sqrt{P_{\text{max}} \cdot R_{\text{Shunt}}} = 11.40\text{V}$$

$$U_{\text{max}} = R_{\text{Shunt}} \cdot I_{\text{Diode_max}} = 65\text{V}$$

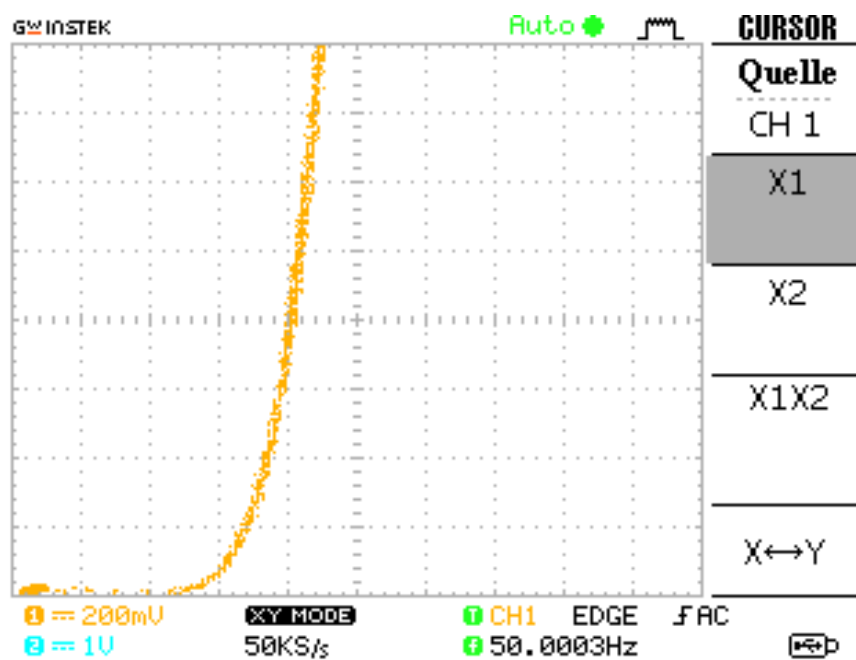
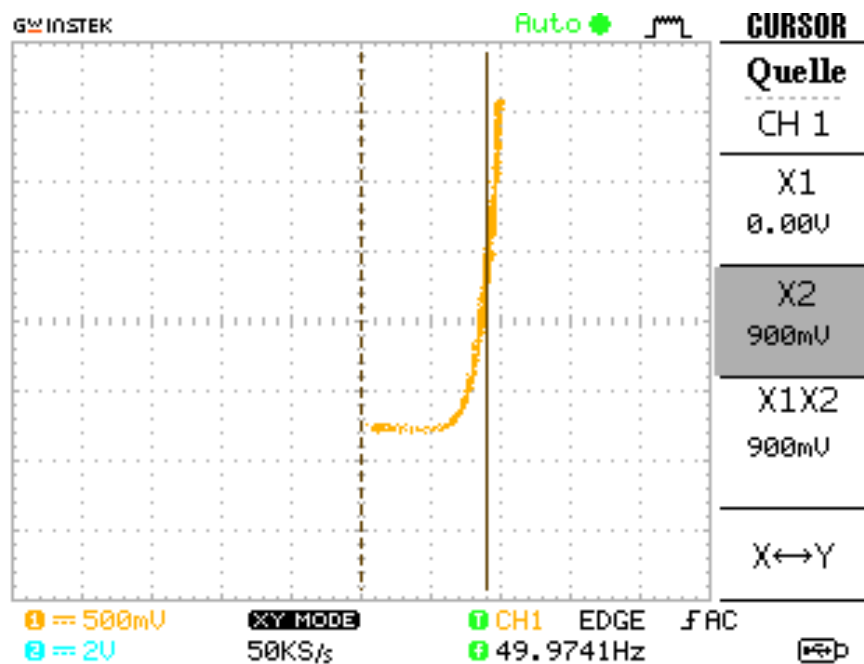
$$U_{\text{Quelle}} = 10\text{V}$$

$$\text{Ch1} = -U_{\text{Diode}}$$

$$\text{Ch2} = I_{\text{ges}}$$

The task was to investigate the behavior of a silicone diode using a oscilloscope and to evaluate the forward threshold voltage U_s .

Measurement:

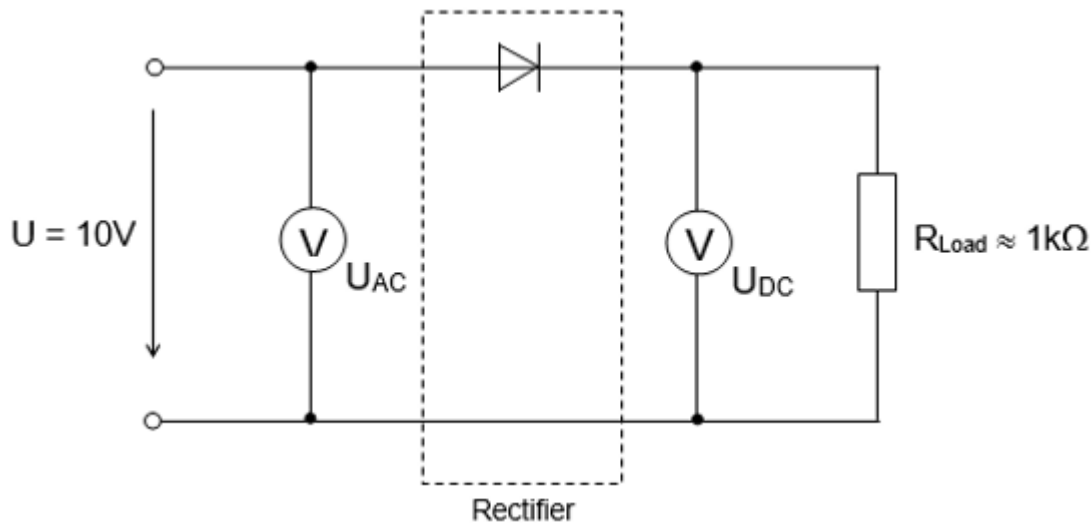


Conclusion:

The picture drawn by the oscilloscope shows a typical silicon diode curve. By using the measurement tools we found out that the forward threshold voltage is roughly 0.9V (on the second picture more like 0.8V). In theory, it should be more like 0.7V, but due to the additional resistance and the not very precise measurement this evaluation is acceptable.

14.2 Half wave rectifier:

Preparation:



U_{AC} -Meter: Fluke 177 True RMS W1.17-73

U_{DC} -Meter: Fluke 177 True RMS W1.17-72

oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17-225

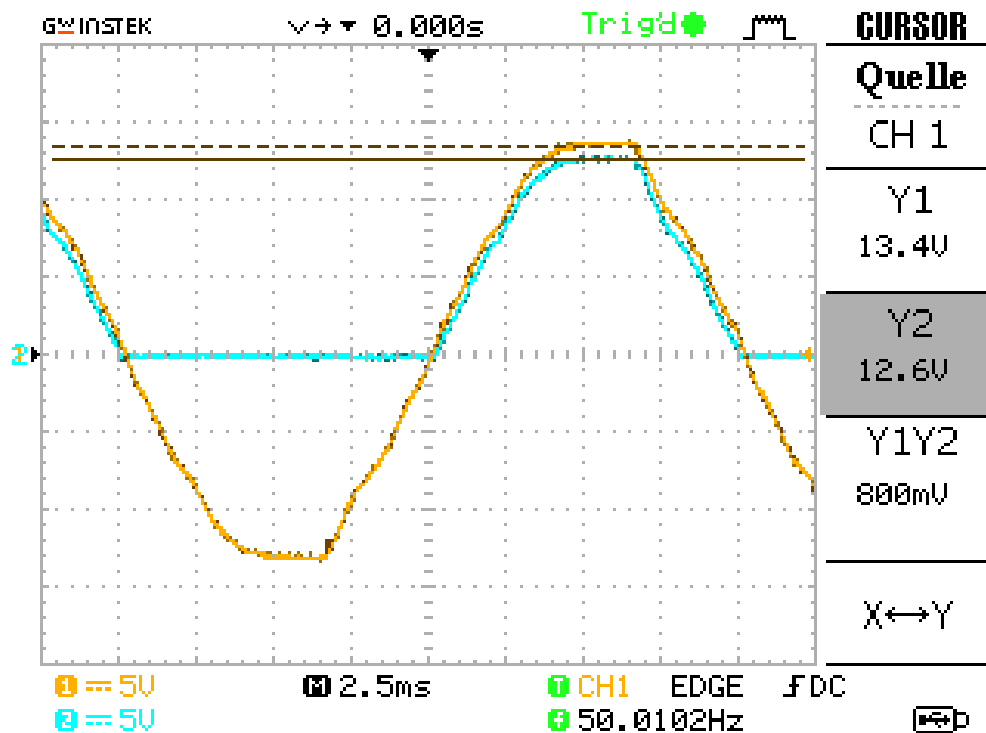
diode: UF5408

$U_{Quelle} = 10V$

The same voltage we measured with the multimeters, should also be measured with the oscilloscope to draw the waveforms. We therefore measured U_{AC} on Channel1 and U_{DC} on Channel2 and connected the ground to the ground of the power supply. For easier reading of the wiring diagram I didn't add the oscilloscopes channel connections.

For this task the voltage before and after our half wave rectifier had to be measured and their waveforms protocolled and compared.

Measurements:



Multimeters:

$U_{AC} = 9.89V$

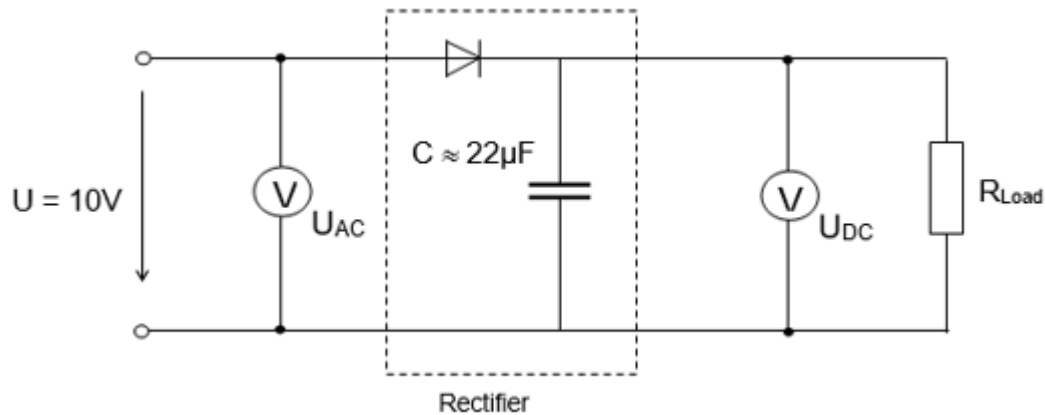
$U_{DC} = 4.81V$

Conclusion:

As expected the AC-voltage has both the positive and negative half waves of the supplied 10V and after our half wave rectifier the negative half waves are being cut off and only the positive ones remain. It can also be seen that at the silicon diode about 0.8V are being lost.

14.3 Half wave rectifier with smoothing capacitor:

Preparation:



U_{AC} -Meter: Fluke 177 True RMS W1.17-73

U_{DC} -Meter: Fluke 177 True RMS W1.17-72

oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17-225

diode: UF5408

$U_{Quelle} = 10V$

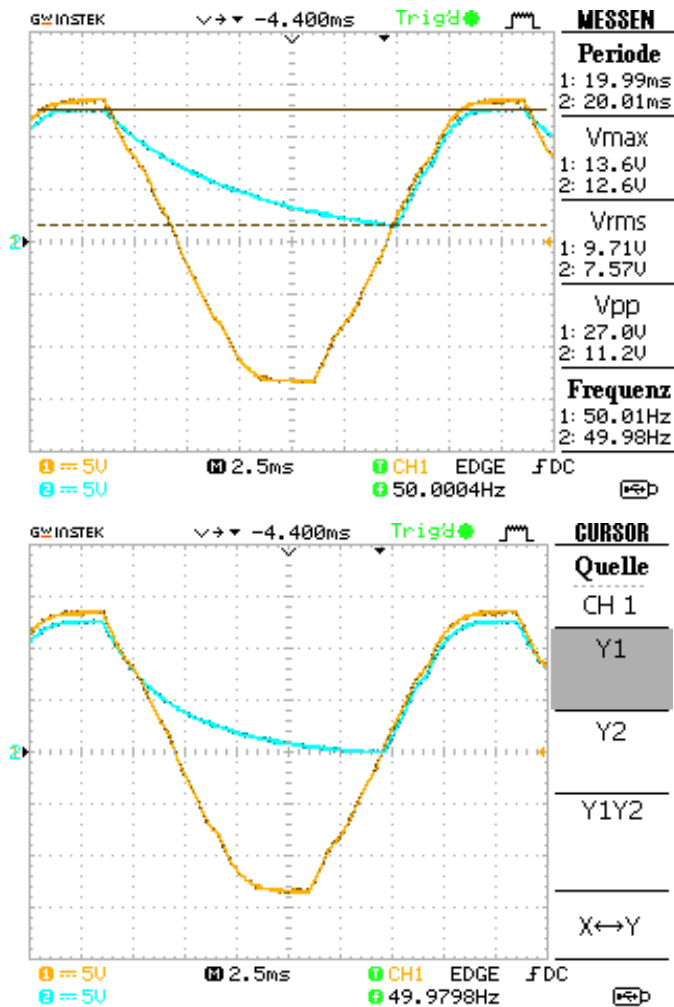
All in all it is the same circuit as in task 14.1, but with an additional smoothing capacitor parallel to our $1k\Omega$ load resistance.

The task was to now additionally investigate the effect of different smoothing capacitors and evaluate their ripple frequency "f" and factor "w". At the end it should also be investigated what affect a reduction of the load resistance to 100Ω has.

$$w = \Delta U_{RMS} / U_{DC}$$

Measurements:

a) 47 μ F



Multimeters: (high load resistance)

$$U_{AC} = 9.90V$$

$$U_{DC} = 6.87V$$

Measurements: (high load resistance)

$$\Delta U_{RMS} = 7.6V$$

$$\Delta U = 11V$$

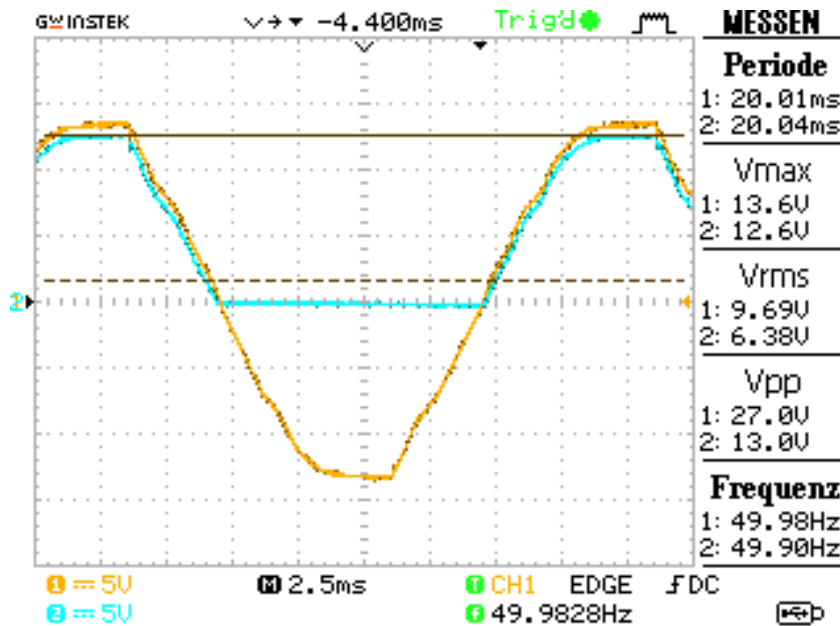
$$w = 1.1$$

Conclusion:

As seen, the 47 μ F smoothing capacitor discharges fairly quickly and is almost completely empty at the begin of the second positive half wave. This discharging increases as the load resistance is decreased too, as the second picture shows. This behaviour is typically for a capacitor as the discharging function is:

$$u(t) = U_0 * e^{-t/R*C}$$

b) 20nF



Multimeters: (high load resistance)

$$U_{AC} = 9.90V$$

$$U_{DC} = 4.81V$$

Measurements: (high load resistance)

$$\Delta U_{RMS} = 6.4V$$

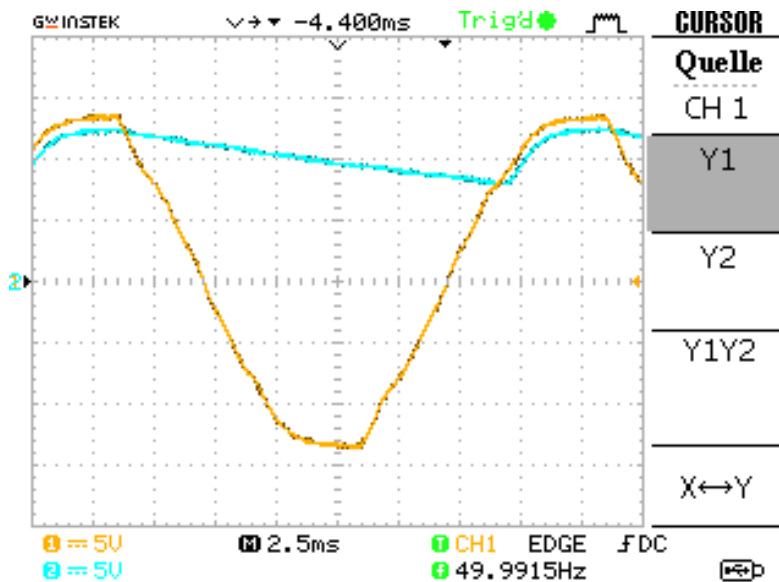
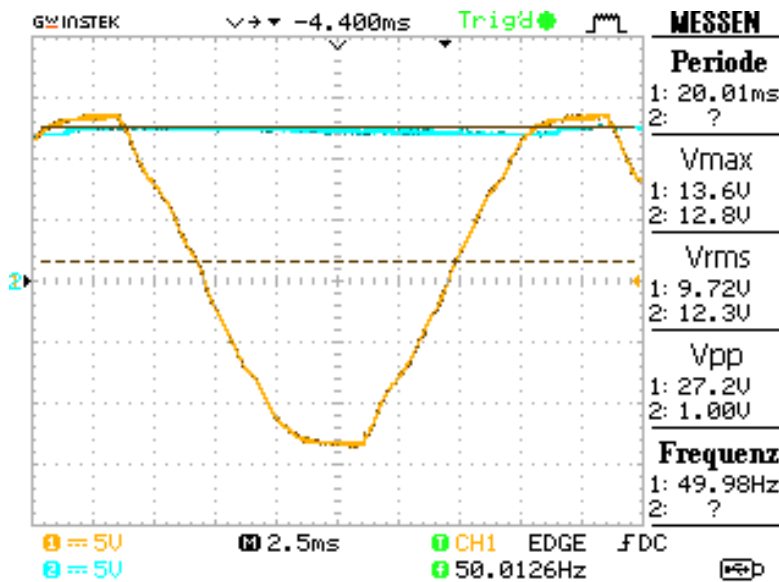
$$\Delta U = 13.2V$$

$$w = 1.53$$

Conclusion:

The 20nF capacitor almost shows, as expected, no “smoothing” at all because the maximum saved energy in this little capacitor isn’t enough to overcome the gap between the two positive half waves.

c) 470 μ F



Multimeters: (high load resistance)

$$U_{AC} = 9.90V$$

$$U_{DC} = 12.6V$$

Measurements: (high load resistance)

$$\Delta U_{RMS} = 12.4V$$

$$\Delta U = 1V$$

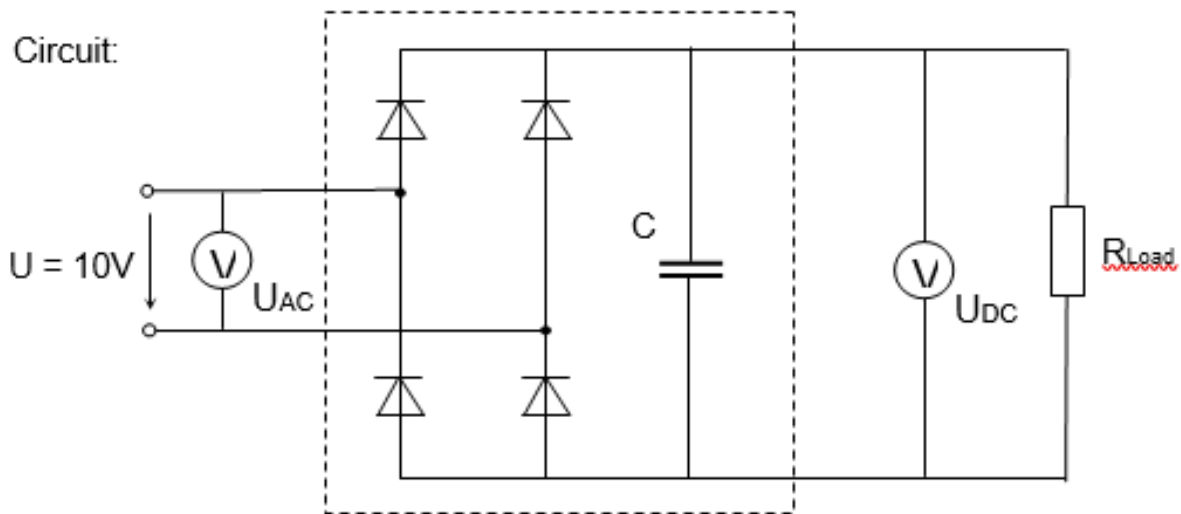
$$w = 0.975 \text{ (should be } >1)$$

Conclusion:

This capacitor would be big enough for a proper half wave rectifier, even with a smaller load resistance as the second picture shows. The behaviour of this capacitor also is as expected and explained in the conclusion section of measurement a).

14.4 Full wave rectifier:

Preparation:



U_{AC} -Meter: Fluke 177 True RMS W1.17-73

U_{DC} -Meter: Fluke 177 True RMS W1.17-72

oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17-225

isolation amplifier: W1.17-705

isolation amplifier: W1.17-703

diodes: UF5408

$U_{Quelle} = 10V$

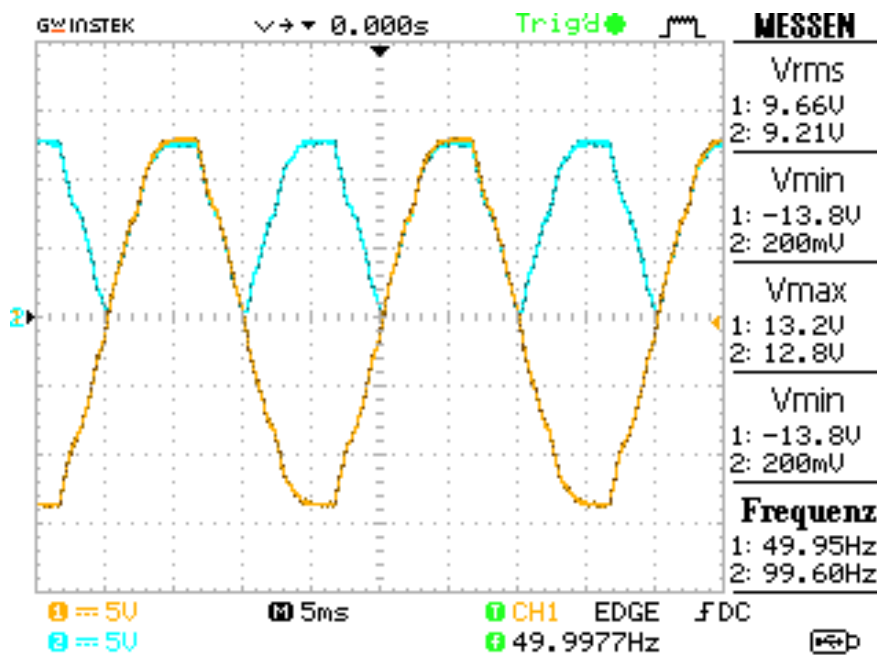
This circuit is very similar to the previous circuit but with four diodes to fully rectify the input AC-signal. This also leads to a ground problem which needs to be solved using isolation amplifiers on both channels.

The task was to investigate the output waveform of the full wave rectifier with the same smoothing capacitors and loads as in task 14.3. Additionally, the waveform of the input current should be measured by using the same R_{Shunt} in series of the input line as in task 14.1.

No oscilloscope wirings are drawn into this wiring diagram for an easier understanding of the circuit.

Measurements:

a) no capacitor



Multimeters: (high load resistance)

$$U_{AC} = 9.83V$$

$$U_{DC} = 7.84V$$

Measurements: (high load resistance)

$$\Delta U_{RMS} = 9.2V$$

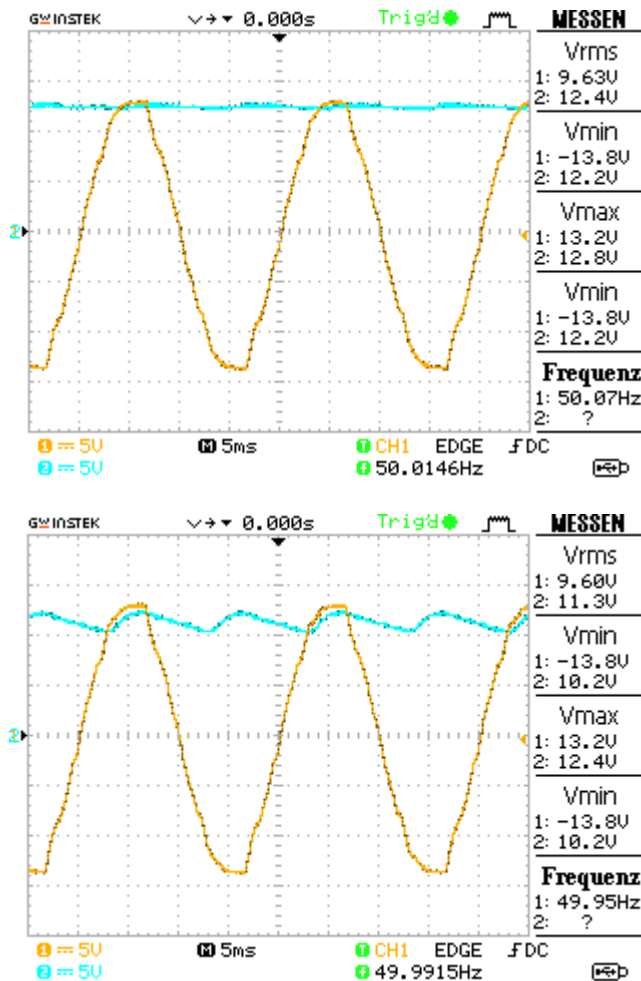
$$\Delta U = 12.8V$$

$$w = 1.17$$

Conclusion:

By using a full wave rectifier the negative half waves of the input signal simple get translated into positive ones, but with a voltage drop of around 0.4V at the diodes as shown in the picture above.

b) 470 μ F



Multimeters: (high load resistance)

$$U_{AC} = 9.95V$$

$$U_{DC} = 12.3V$$

Measurements: (high load resistance)

$$\Delta U_{RMS} = 12.4V$$

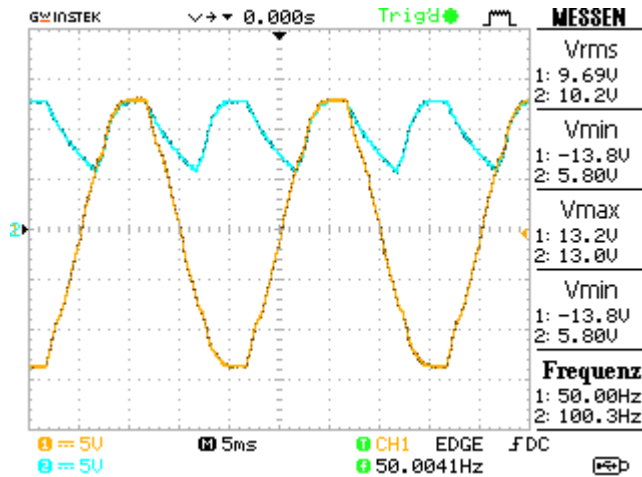
$$\Delta U = 0.6V$$

$$w = 1$$

Conclusion:

The lower picture again shows the faster discharge of the smoothing capacitor when paired with a low load resistance, although the effect isn't as dramatically as on the half wave rectifier because the capacitor now also utilizes the rectified negative half waves to charge up.

c) 47 μ F



Multimeters: (high load resistance)

$$U_{AC} = 9.99V$$

$$U_{DC} = 9.67V$$

Measurements: (high load resistance)

$$\Delta U_{RMS} = 10.2V$$

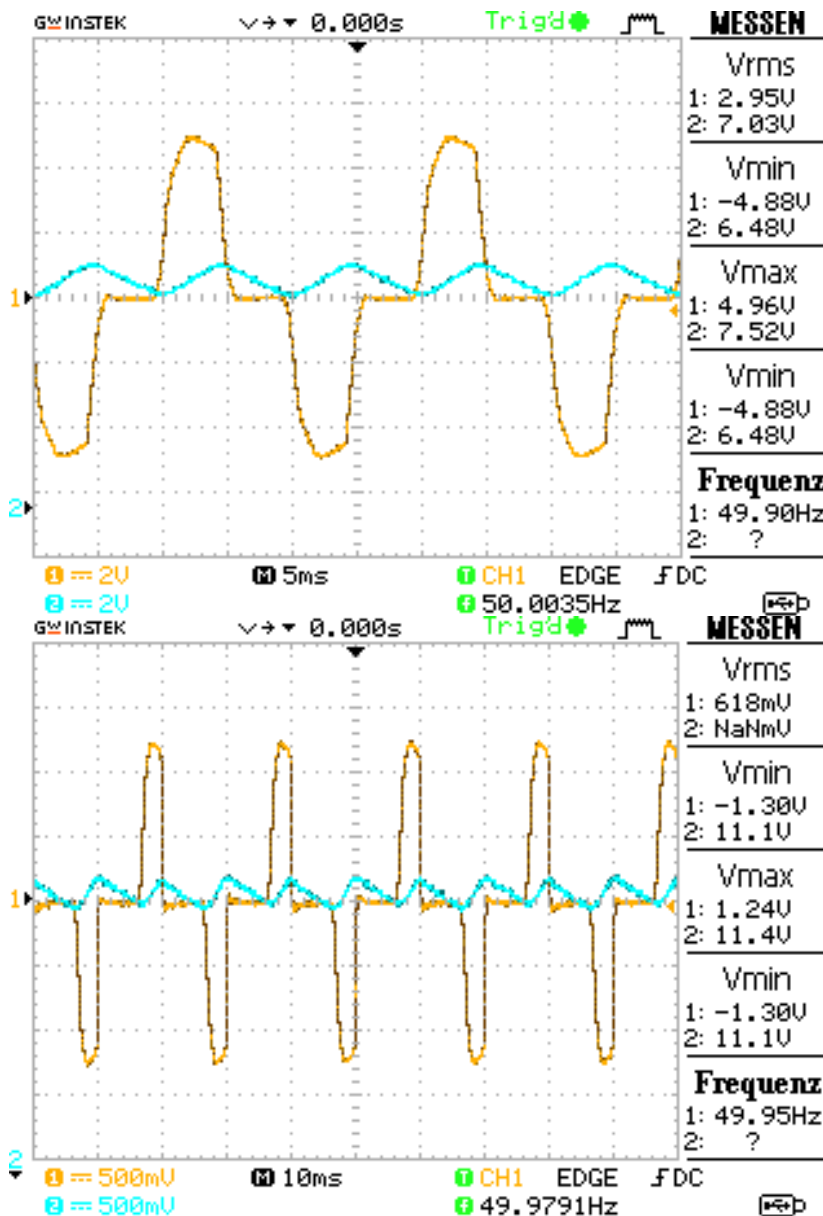
$$\Delta U = 7.2V$$

$$w = 1.05$$

Conclusion:

This evaluation proves the improved efficiency of the full wave rectifier as the ripple factor decreased to 1.05 from 1.1 of the half wave rectifier. Now the voltage also never drops to 0V because of the added charge cycle of the rectified negative half waves.

d) currents



Conclusion:

The orange graph shows the current draw of the system, the blue graph the input AC-voltage shifted upwards on the y-scale for a better visualization.

The upper picture shows a full wave rectifier with a 470 μ F capacitor paired with a 1k Ω load resistance, the lower picture the same capacitor but with a decreased load resistance of just 100 Ω .

Here it can be seen that the faster discharge of the capacitor because of the lower load resistance leads to a lower peak current since the capacitor has more time to charge up. The gaps between these current peaks show that the capacitor is discharging and therefore powering the system without any input current.